

```

import pandas as pd
import numpy as np

df = pd.read_csv('/content/covid_toy.csv')

df.head()

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 100,\n  \"fields\": [\n    {\n      \"column\": \"age\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 24,\n        \"min\": 5,\n        \"max\": 84,\n        \"num_unique_values\": 55,\n        \"samples\": [\n          55,\n          84,\n          49\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"gender\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Female\",\n          \"Male\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"fever\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2.0549261782367014,\n        \"min\": 98.0,\n        \"max\": 104.0,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          103.0,\n          100.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"cough\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Strong\",\n          \"Mild\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"city\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 4,\n        \"samples\": [\n          \"Delhi\",\n          \"Bangalore\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"has_covid\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Yes\",\n          \"No\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}, \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

age is continuous data, gender is categorical data, fever is continuous data, cough is categorical city is categorical data -> nominal, has_covid is categorical -> nominal data target is label encoder, has_covid (label encoder)

```

# can see unique data
df['cough'].unique()

array(['Mild', 'Strong'], dtype=object)

df.isnull().sum() # checking is missing value

```

```
age          0
gender       0
fever        10
cough        0
city         0
has_covid    0
dtype: int64
```

```
df.head()
```

```
{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 100,\n  \"fields\": [\n    {\n      \"column\": \"age\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 24,\n        \"min\": 5,\n        \"max\": 84,\n        \"num_unique_values\": 55,\n        \"samples\": [\n          55,\n          84,\n          49\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"gender\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Female\",\n          \"Male\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"fever\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2.0549261782367014,\n        \"min\": 98.0,\n        \"max\": 104.0,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          103.0,\n          100.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"cough\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Strong\",\n          \"Mild\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"city\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 4,\n        \"samples\": [\n          \"Delhi\",\n          \"Bangalore\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"has_covid\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Yes\",\n          \"No\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}, \"type\": \"dataframe\", \"variable_name\": \"df\"}
```

```
df.shape
```

```
(100, 6)
```

```
from sklearn.model_selection import train_test_split
```

```
x = df.drop(columns=['has_covid'])
y = df['has_covid']
```

```
X_train, X_test, Y_train, Y_test = train_test_split(x, y, test_size= 0.2,
random_state=40)
```

random_state can not change data sequence

X_train

```
{
  "summary": {
    "name": "X_train",
    "rows": 80,
    "fields": [
      {
        "column": "age",
        "properties": {
          "dtype": "number",
          "std": 25,
          "min": 5,
          "max": 84,
          "num_unique_values": 46,
          "samples": [
            15,
            20,
            69
          ],
          "semantic_type": ""
        }
      },
      {
        "column": "gender",
        "properties": {
          "dtype": "category",
          "num_unique_values": 2,
          "samples": [
            "Male",
            "Female"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "fever",
        "properties": {
          "dtype": "number",
          "std": 2.0371294706844876,
          "min": 98.0,
          "max": 104.0,
          "num_unique_values": 7,
          "samples": [
            101.0,
            104.0
          ],
          "semantic_type": ""
        }
      },
      {
        "column": "cough",
        "properties": {
          "dtype": "category",
          "num_unique_values": 2,
          "samples": [
            "Strong",
            "Mild"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "city",
        "properties": {
          "dtype": "category",
          "num_unique_values": 4,
          "samples": [
            "Delhi",
            "Mumbai"
          ],
          "semantic_type": ""
        }
      }
    ],
    "description": ""
  },
  "type": "dataframe",
  "variable_name": "X_train"
}
```

X_train.shape

```
(80, 5)
```

X_test.shape

```
(20, 5)
```

Y_train.shape

```
(80,)
```